

Name:

Solutions

Date:

11/08/2015

Year 10A Mathematics Course

Test 4, 2015

Topics – Pythagoras and Trigonometry

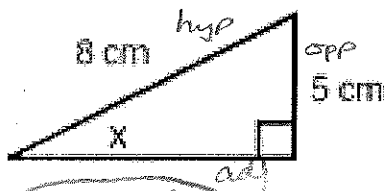
46

= %

Total Time: **60** minutesTotal Reading: **5** minutesTotal Working: **55** minutesWeighting: **5 %** of the year.Equipment Allowed: **1 page notes (A4 both sides), Scientific calculator**

CALCULATOR ASSUMED: Show all working for full marks, place answer in space if provided.

1. [2 marks]

Find the angle x , to 1 d.p.

X=

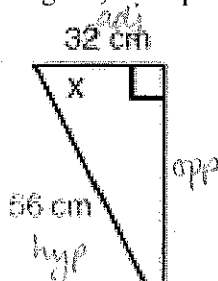
38.7°

$$\begin{aligned} \text{opp} &= 5 \\ \text{hyp} &= 8 \\ \text{angle} &= ? \end{aligned}$$

$$\sin x = \frac{o}{h}$$

$$\begin{aligned} \therefore x &= \sin^{-1}\left(\frac{5}{8}\right) \\ &= 38.682 \end{aligned}$$

2. [2 marks]

Find the angle x , to 1 d.p..

X=

55.2°

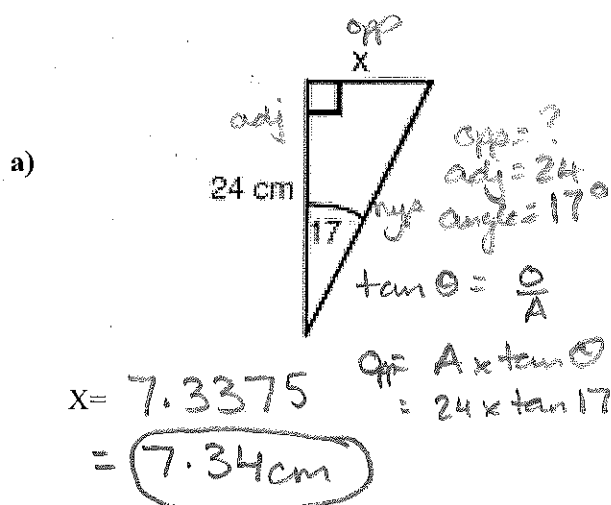
$$\begin{aligned} \text{adj} &= 32 \\ \text{hyp} &= 56 \\ \text{angle} &= ? \end{aligned}$$

$$\cos x = \frac{a}{h}$$

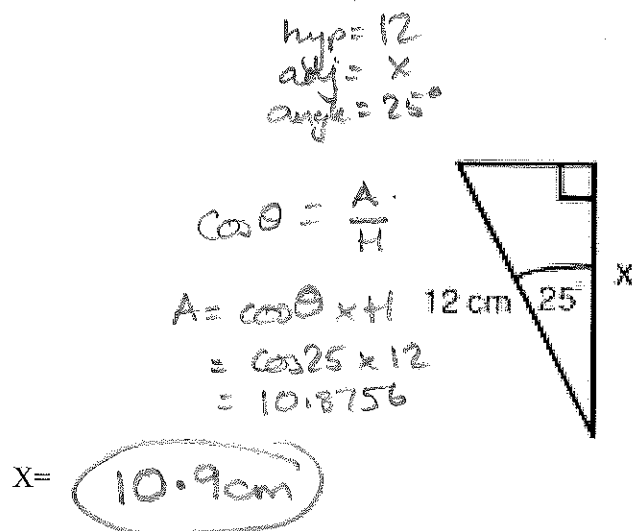
$$\begin{aligned} \therefore x &= \cos^{-1}\left(\frac{32}{56}\right) \\ &= 55.15009 \end{aligned}$$

3. [4 marks: 2, 2]

Find the side lengths marked x, to 3 significant figures.



b)



4. [4 marks: 2, 2]

A ladder that is 6 metres long is placed against a wall.

It makes an angle of 34° with the wall. Find

a). how high up the wall it reaches,

b). the distance the base of the ladder is away from the wall.

$\cos \theta \times H = \text{Adj}$
 $\cos 34^\circ \times 6 = 4.974$

$\sqrt{(6^2 - 4.974^2)}$
 $= 3.36 \text{ m}$
or
 $\sin 34^\circ \times 6$
 $= 3.36 \text{ m}$

a) 4.97m

b) 3.36m

Height = 4.97m

Base distance from wall = 3.36m

5. [5 marks: 2, 3.]

(a) Find the value of x, show working:

$\frac{x}{\sin 40} = \frac{6}{\sin 30}$

$x = \frac{6 \times \sin 40}{\sin 30}$
 $= 7.71$



(b) Find the missing angle and hence find the area.

$A = \frac{1}{2} ab \sin c = \frac{1}{2} \times 7.71 \times 6 \times \sin 110$

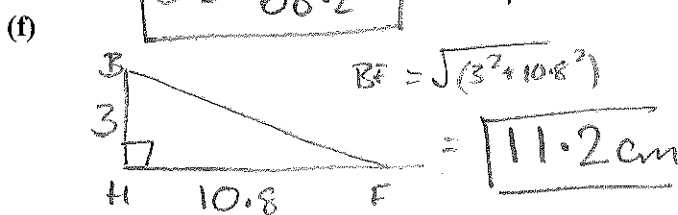
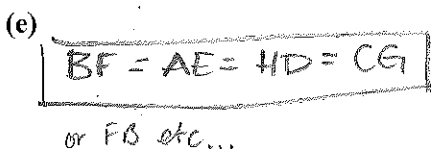
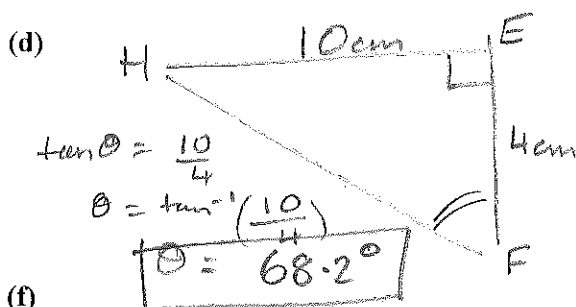
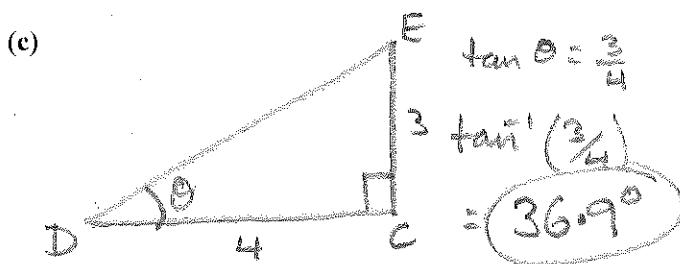
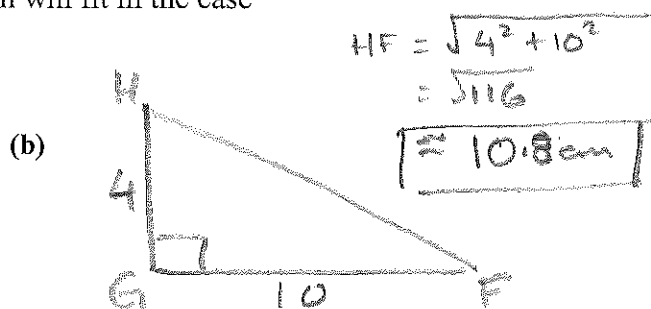
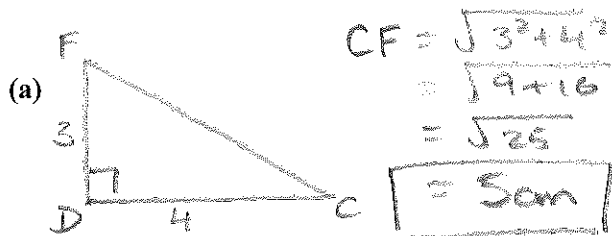
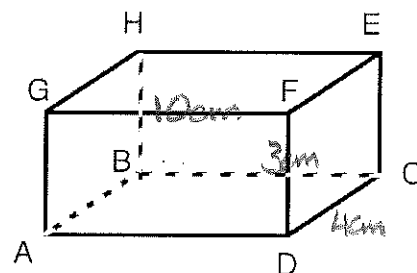
Angle = 110° degrees.

Area = 21.7 units squared.

6. [10 marks: 1, 1, 2, 2, 2, 2]

In the cuboid $CD = 4$ cm, $DF = 3$ cm, $FG = 10$ cm. Find

- the diagonal CF ,
- the diagonal HF ,
- $\angle CDE$,
- $\angle EFH$.
- If this was a pencil case identify the longest distances possible inside the case ?
- Hence, find the length of the longest pencil which will fit in the case

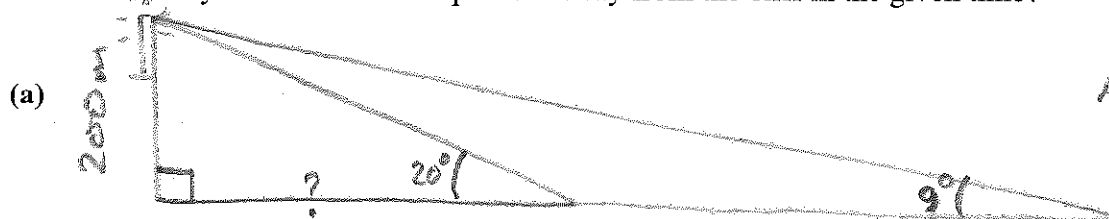


7. [6 marks: 4, 2]

A lighthouse 20 m tall, is on top of cliffs 180 m high.

At 9 a.m. the angle of depression of a ship from the top of the lighthouse was 20° and at 9.10 a.m. it was 9° . Draw a diagram of this below, use a straight edge or ruler.

How many metres did the ship travel away from the cliff in the given time?



At 0900 $? = \frac{200}{\tan 20^\circ} = 549.5 \text{ m}$
At 0910 $? = \frac{200}{\tan 9^\circ} = 1262.7$

What was the ship's average speed in km/h? Give your answer to two decimal places. DISTANCE gone = 713 m

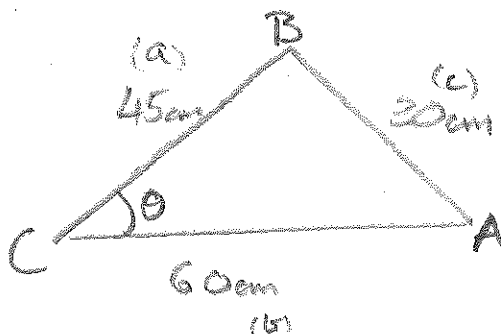


8. [5 marks 3, 2.] ---

If $a = 45\text{cm}$, $b = 60\text{cm}$, and $c = 30\text{cm}$,

(a) What is the value of the smallest angle θ in the triangle ABC.

Give your answer correct to one decimal place. (Hint: you may prefer to draw a diagram)



$$\theta = 29^\circ$$

$$\cos \theta = \frac{(45^2 + 60^2) - 30^2}{2 \times 45 \times 60}$$

$$\begin{aligned} \therefore \theta &= \cos^{-1} \left(\frac{(45^2 + 60^2) - 30^2}{2 \times 45 \times 60} \right) \\ &= \cos^{-1} \left(\frac{4725}{5400} \right) \\ &= 28.955 \\ &\approx 29.0^\circ \end{aligned}$$

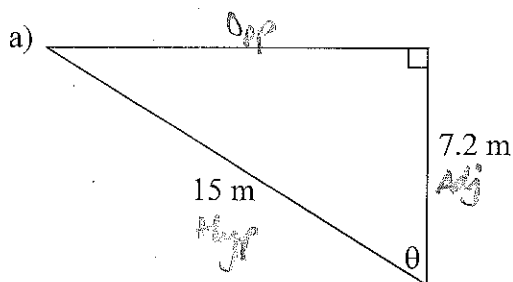
(b) What is the area of this triangle ABC, to the nearest square centimetre?

$$\begin{aligned} \text{Area} &= 0.5 \times 45 \times 60 \times \sin 29^\circ \\ &= 654.49 \\ &\approx 654 \text{ cm}^2 \end{aligned}$$

$$\text{Area} = 654 \text{ cm}^2$$

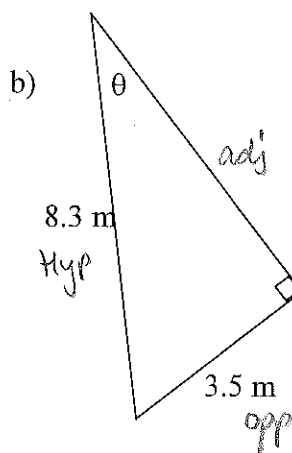
9. [4 marks 2, 2.]

For the following triangles, find the missing side and hence, or otherwise, find the angle marked θ . (Hint: mark the sides with Hyp, Opp and Adj)



$$\cos \theta = \frac{7.2}{15}$$

$$\begin{aligned} \theta &= \cos^{-1} \left(\frac{7.2}{15} \right) \\ &= \underline{\underline{61.3^\circ}} \end{aligned}$$



$$\sin \theta = \frac{3.5}{8.3}$$

$$\theta = \sin^{-1} \left(\frac{3.5}{8.3} \right)$$

$$\theta = \underline{\underline{24.9^\circ}}$$

10. [4 marks 2, 2.]

Using Pythagoras' Theorem or otherwise, state whether or not the lengths make a right-angled triangle. Show your working.

a) [9, 40, 41]

If 90° ; then,

$$41^2 = 40^2 + 9^2$$

$$1681 = 1600 + 81 \quad \text{TRUE}$$

∴ 90° and
right angled
triangle

b) [5, 17, 18]

$$18^2 = 5^2 + 17^2$$

$$324 = 25 + 289$$

$$324 \neq 314$$

∴ FALSE and not a 90°

triangle as
Pythagoras Theorem
does not apply.

~ END OF TEST ~